

Javier García Gilabert

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Education



Master's Exchange Student (Erasmus)

AALTO UNIVERSITY

Helsinki, FI

Sep. 2023 - Jan. 2024

- Got the best Exchange student award in my master's program.



Master in Artificial Intelligence

POLYTECHNIC UNIVERSITY OF CATALONIA (UPC) & UNIVERSITY OF BARCELONA (UB) & UNIVERSITY OF ROVIRA I VIRGILI (URV)

Barcelona, ES

2022 - 2024



- Interuniversity master (taught in English).
- Achieved a mean degree mark of 8.85/10.
- Master Thesis: *Improving Multilingual Neural Machine Translation by projecting language representations.*



Bachelor's degree in Data Science

POLYTECHNIC UNIVERSITY OF VALENCIA (UPV)

Valencia, ES

2018 - 2022

- Got the best student award in Data Science. Honor granted for achieving the highest marks in my university degree.
- Achieved a mean degree mark of 9.3/10.
- Subjects passed with honors: 13.
- Received a Special Mention for my Bachelor Thesis: *Image Captioning using pre-trained GPT-2 models.*

Experience



Research Engineer in NLP

BARCELONA SUPERCOMPUTING CENTER

Barcelona, ES

Jan. 2024 - Present

Working as a research engineer in the Language Technologies Group at the Barcelona Supercomputing Center.

- NLP research engineer in Eloquence (European Project), ILENIA (Spanish National Project), AINA (Catalan Project).
- Developed Multilingual Neural Machine Translation models LLM-based (SalamandraTA-2b-instruct, SalamandraTA-7b-instruct, Plume).
- Worked in low-resource machine translation.
- Experience in training LLMs in multi-node distributed environments using DeepSpeed and Nemo frameworks.



Research Engineer in NLP

MT@UPC (MACHINE TRANSLATION GROUP AT UPC)

Barcelona, ES

Oct. 2022 - Sep. 2023

National Spanish Project ROB-IN (PLEC2021-007859). The primary goal of this project was to create a robot designed for continuous personalized assistance with self-explanatory capabilities. My role in this project involved contributing to the development of the Natural Language Processing (NLP) component. Duties included:

- Developing the dialogue system of the robot using the RASA framework.
- Investigating the phenomenon of hallucinations in neural machine translation models.



Research Collaboration Fellowship

PRHLT (PATTERN RECOGNITION AND HUMAN LANGUAGE TECHNOLOGY RESEARCH CENTER)

Valencia, ES

Oct. 2021 - Aug. 2022

I received a collaboration fellowship with the PRHLT group from the Spanish Ministry of Education, where I focused on researching state-of-the-art encoder-decoder architectures for image captioning. This research led to the development of my bachelor's thesis under the supervision of Francisco Casacuberta Nolla.



Undergraduate Research Assistant

DEPARTMENT OF APPLIED STATISTICS, OPERATIONS RESEARCH AND QUALITY AT UPV

Valencia, ES

Oct. 2019 - Aug. 2022

My role was mainly as a data scientist, dealing with data collection, processing, analysis, and the application of machine learning techniques.

Publications



Javier Garcia Gilabert, Carlos Escolano, Xixian Liao and Maite Melero. 2025. Terminology-Constrained Translation from Monolingual Data using GRPO. *In Proceedings of the Tenth Conference on Machine Translation*.



Javier Garcia Gilabert, Xixian Liao, Severino Da Dalt, Ella Bohman, Audrey Mash, Francesca De Luca Fornaciari, Irene Baucells, Joan Llop, Miguel Claramunt Argote, Carlos Escolano and Maite Melero. 2025. From Salamandra to SalamandraTA: BSC Submission for WMT25 General Machine Translation Shared Task. *In Proceedings of the Tenth Conference on Machine Translation*.



Javier Garcia Gilabert, Carlos Escolano, Aleix Sant, Francesca De Luca Fornaciari, Audrey Mash, Xixian Liao, and Maite Melero. 2024. Investigating the translation capabilities of Large Language Models trained on parallel data only. *In Proceedings of Machine Translation Summit XX: Volume 1, pages 24–53, Geneva, Switzerland. European Association for Machine Translation*.



Javier Garcia Gilabert, Carlos Escolano, Audrey Mash, Xixian Liao, and Maite Melero. 2024. MT-LENS: An all-in-one Toolkit for Better Machine Translation Evaluation. *In Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (System Demonstrations), pages 51–60, Albuquerque, New Mexico. Association for Computational Linguistics*.



Javier Garcia Gilabert, Aleix Sant, Carlos Escolano, Francesca De Luca Fornaciari, Audrey Mash, and Maite Melero. 2024. BSC Submission to the AmericasNLP 2024 Shared Task. *In Proceedings of the 4th Workshop on Natural Language Processing for Indigenous Languages of the Americas (AmericasNLP 2024), pages 143–149, Mexico City, Mexico. Association for Computational Linguistics*.



Javier Garcia Gilabert, Carlos Escolano, and Marta R. Costa-Jussà. 2024. ReSeTOX: Re-learning attention weights for toxicity mitigation in machine translation. *In Proceedings of the 25th Annual Conference of the European Association for Machine Translation, pages 37–58, Sheffield, England. European Association for Machine Translation*.

Peer collaborations:



Aleix Sant, Daniel Bardanca, José Ramon Pichel Campos, Francesca De Luca Fornaciari, Carlos Escolano, **Javier Garcia Gilabert**, Pablo Gamallo, Audrey Mash, Xixian Liao and Maite Melero. 2024. Training and Fine-Tuning NMT Models for Low-Resource Languages using Apertium-Based Synthetic Corpora. *In Proceedings of the Ninth Conference on Machine Translation, pages 925–933, Miami, Florida, USA. Association for Computational Linguistics*.



Xixian Liao, Carlos Escolano, Audrey Mash, Francesca De Luca Fornaciari, **Javier García Gilabert**, Miguel Claramunt Argote, Ella Bohman and Maite Melero. 2025. Culture-aware machine translation: the case study of low-resource language pair Catalan-Chinese. *In Proceedings of Machine Translation Summit XX: Volume 1, pages 150–161, Geneva, Switzerland. European Association for Machine Translation*.

Awards and Grants

2024	Best Exchange Student award in Master in Artificial Intelligence Honor granted for achieving the highest marks in my master's exchange program.	UPC-URV-UB
2022	Best Student award in Data Science Honor granted for achieving the highest marks in my university degree.	UPV
2022	Beca Santander Estudios Progreso 2021-2022 The "Becas Santander Progreso" program is a grant given by Banco Santander S.A., whose objective is to facilitate access to education for students with the best academic records at a national level who are pursuing official Bachelor's or Master's degree studies.	Banco Santander
2021	Awards to Graduates of the UPV with Best Academic Records Honor granted for achieving the highest marks in the second academic year of my University degree.	UPV

Service in the field

2025	Reviewer EMNLP Demo Track 2025
2025	Reviewer WMT 2025
2025	Reviewer MT Summit 2025
2025	Reviewer NAACL Demo track 2025

Certificates

2023 **DeepLearn 2023 summer: 10th International Gran Canaria School on Deep Learning**
2017 **First Certificate of English (FCE)**

*DeepLearn
Cambridge*

Skills

Programming Python, R, LaTeX, Bash.
Frameworks PyTorch, numpy, pandas, nemo, deepspeed, slurm, gradio, vllm, fastapi, lm-evaluation-harness, accelerate, transformers.
Languages Spanish (Native), Catalan (Native), English (Fluent).

Volunteering Activities

Mentor for international exchange students

UPV

MENTOR

2022

Assisting international exchange students in their adaptation to the university and the city of Valencia. As part of the university's

MENTOR support program: <https://www.upv.es/entidades/OPII/infoweb/pi/info/820230normali.html>

Mentor for new university students

UPV

PATU

2019-2021

Advising new Data Science students in their transition to the university, as part of PATU (Programa de Acción Tutorial):

<https://www.upv.es/perfiles/futuro-alumno/integra-patu-es.html>